Computer Support Specialist Mini Project 3

Network Designing and Troubleshooting

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Introduction

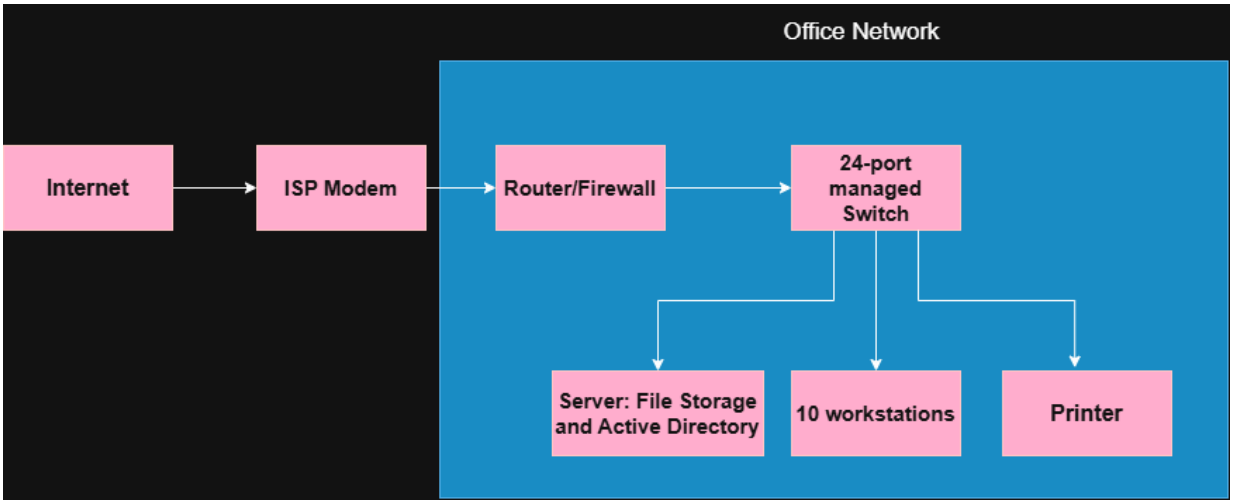
This project designs a network for a small office with 10 users. It describes the network devices and how they connect, and explains how the design supports security, reliability and future growth. The project also includes a plan for troubleshooting a network problem.

Scenario 1: Network Design for a Small Office

Scenario Understanding

This is a small office at one location with 10 users. The network needs to support 10 workstations, one server for file storage and Active Directory, one shared printer, and internet access. Users need reliable access to files and printing during business hours. The design should be cost-conscious, allow some future growth, and include basic security.

Network Diagram



Network Components and Configuration

Router:

The router connects the office network to the ISP modem and provides internet access. Its firewall is enabled to help block unauthorized network traffic.

Switch:

I chose a managed switch because it connects the office devices and supports separate network groups. This helps keep the server network logically separate from the users' workstations.

Server:

The server stores shared files and runs Active Directory to manage user accounts and their access to network resources. It has a fixed IP address so the workstations can find it reliably.

Printer:
The multi-function printer connects to the switch with an Ethernet cable. It is shared by all users, and it has a reserved IP address so their workstations can find it consistently.
Workstations:
The 10 workstations connect to the switch using Ethernet cables. They receive their network settings automatically through DHCP and can access the internet, shared files and printer. Their users sign in with accounts managed by Active Directory.

Design Decisions

Network Topology

Decision:
Star topology using a central managed switch.
Justification:
Each device connects to the central switch. This makes the network easy to manage and troubleshoot. If one cable fails, it should affect only the device connected to that cable, which suits a small office with 10 users.

IP Addressing Approach

Decision:
I will use a mix of DHCP and fixed addresses.
Justification:
DHCP will automatically give network settings to the workstations, so they do not need to be set up one by one. The server will have a fixed IP address so users can reliably find it. The printer will have a reserved IP address so it keeps the same address for users.

Security Baseline
The router/firewall is placed between the ISP modem and the office network, and its firewall is enabled. I will use separate VLANs to keep the server and workstations in different network groups. Firewall rules will allow workstations to use the server services they need while blocking unnecessary traffic.
Design Tradeoff
To keep costs low, the office will use one ISP connection instead of paying for a second internet connection. If the ISP connection fails, users will temporarily lose internet access, although local file sharing and printing may still work. For urgent internet tasks, staff can use a mobile hotspot until the ISP service is restored. This is an acceptable tradeoff for a small office with 10 users.
Scalability, Security, and Performance

- **Future growth:** A 24-port managed switch leaves spare ports for additional devices.
- **Basic security:** The enabled firewall and separate VLANs help protect the server and limit unnecessary network traffic.
- **Reliable performance:** Wired connections give the workstations, server and printer a stable connection during business hours.

Scenario 2: Troubleshooting Network Usability Issues

Reported Symptoms

Some users report that shared files are slow to open. Other users cannot connect to certain network services. The issue affects some users or services, rather than stopping the whole network. The affected users, services and times of the problem need to be confirmed.

Clarifying Questions

1. Which users or workstations are affected?
2. Which shared files or network services are slow or unavailable?
3. When does the problem happen, and how long does it last?
4. Can affected users still access the internet and the printer?

Possible Root Causes

1. **Incorrect DNS settings on some workstations** may stop them from finding the server or network services by name.
2. **A server performance problem**, such as a busy or slow disk, may make shared files slow to open.

Ruled-Out Root Cause

I ruled out the **server disk bottleneck**

Explanation:

I ruled out a server disk bottleneck because the server's CPU, memory and disk activity were normal during the problem. The shared files opened quickly when I accessed them using the server's IP address, so the server disk was unlikely to be the cause. This pointed towards a problem finding the server by name.

Troubleshooting Methodology

I first identified which users and services were affected. I checked the network settings on an affected workstation and compared them with a working one. I then tested access to the server by its name and by its IP address. Access worked by IP address, but not reliably by name. I also checked the server's activity and found no sign of a disk or resource problem. These checks pointed to a DNS settings problem.

Final Diagnosis

The most likely root cause is incorrect DNS settings on some workstations, making it difficult for them to find the server and network services by name.

Proposed Solution

I would correct the DHCP settings so affected workstations use the office server's DNS service to find network resources by name. Then I would renew the workstations' network settings and test access to the shared files and services. This solution addresses the DNS problem shown by the tests. I chose it over replacing the server or switch because those devices were working normally.

Impact on Usability and Reliability

Users should be able to open shared files and connect to network services by name more quickly and consistently. Correct DHCP settings also help workstations receive the right DNS information, reducing the chance of the same problem happening again.

Optional Recording

Watch my presentation video Presentation Video.mp4